

Community Restoration Platform

Final MVP Product & Technical Documentation — Draft for Review

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Core philosophy: **Plant it. Record it. Monitor it. Learn from it.**

This document defines the proposed Minimum Viable Product (MVP) for a community-driven ecosystem restoration platform. The product supports both direct seeding and sapling plantation, records real-world activities, enables community participation, and creates a foundation for future ecological analysis and expert guidance.

1. Product Definition

The platform is a community-driven restoration system where people can seed, plant, organize restoration activities, join other activities, record location and evidence, monitor outcomes, and view community progress.

The MVP does not promise a specific number of surviving trees or a specific climate impact. Its primary purpose is to establish the community, create structured restoration records, and collect real-world data that can later be used to determine what methods work best in different environments.

2. Product Principles

Principle	Meaning
Activity $\neq$ Success	A reported seed or planted sapling is recorded as an activity, not automatically as a successful plant.
Failure is Data	Failed germination or plant mortality is still useful information for future analysis.
Two Restoration Methods	Seeding and plantation are supported separately because their processes and outcomes differ.
Quantity Is Not Everything	Leaderboards should not reward raw quantity at the expense of monitoring and quality.
Evidence Increases Confidence	GPS, timestamp, photos, monitoring, and later verification progressively increase confidence in a record.
Expertise Comes Later	The MVP creates a structure that experts, researchers, and organizations can improve in future phases.

3. Core Concepts

Restoration Activity — A single recorded real-world action. MVP types: SEEDING and PLANTATION.

Restoration Site — A geographic place or area where one or more restoration activities occur.

Monitoring Record — A later observation about an activity or site, such as observed seedlings, established plants, survivors, dead plants, or unknown status.

Community Task — An organized restoration activity created by a user for other community members to join.

4. User Roles

Role	MVP Capabilities
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User	Create seeding/plantation activities, join tasks, monitor activities, view map, goals, leaderboards and profile.
Admin	Manage users, species, activities, sites, goals, reports, moderation and basic analytics.

5. Core User Journey

Register → Explore Map → Create or Join Activity → Seed or Plant → Record Location + Evidence → Activity Created → Monitor Later → Report Outcome → Community Statistics

6. Seeding Activity

- **Species:** selected from the supported species list.
- **Quantity:** estimated number of seeds deployed.
- **Method:** direct seeding or another supported seeding method.
- **Location:** GPS coordinates, optional manual map selection, GPS accuracy, date/time.
- **Evidence:** photos and optionally a short video.
- **Notes:** optional observations such as terrain, rainfall, soil condition, or local context.

7. Plantation Activity

- **Species**
- **Number of saplings**
- **Plantation method**
- **Location and GPS accuracy**
- **Date/time**
- **Photos**
- **Optional notes**

8. Map & Restoration Sites

The map is a core MVP feature. Users can view restoration activities and restoration sites geographically. A site can contain multiple activities and contributors. The system should not attempt to track individual seeds or individual plants by GPS.

Important: GPS proves that a device was at or near a location; it does not prove that a particular number of seeds was actually deployed. Therefore the platform should use terms such as **Reported**, **Evidence Provided**, **Verified**, and **Monitored** rather than claiming absolute proof.

9. Community Tasks

Users can create organized activities such as a weekend mountain seeding event or a community plantation event. Other users can join and contribute. For the MVP this should be treated as community participation/volunteering, not a paid hiring marketplace.

10. Monitoring System

Monitoring is a core differentiator. A restoration activity may later receive one or more monitoring records.

Stage	Example
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Initial activity	1,000 seeds reported
Monitoring 1	240 seedlings observed
Monitoring 2	180 established
Monitoring 3	120 surviving

11. Activity & Outcome Status

Activity: Reported → Evidence Provided → Verified → Monitored. Outcome observations may be recorded as Observed, Established, Surviving, Dead, or Unknown. The exact ecological definitions should be reviewed by a qualified restoration/forestry expert before production use.

12. Historical Survival Estimates

The proposed statistics feature may eventually estimate expected survivors using historical platform observations. For example, if a sufficiently large and comparable dataset shows a 27% observed survival rate, 1,000 reported seeds could display **approximately 270 estimated survivors**. This must be explicitly labelled as an estimate, not a guarantee.

MVP rule: Do not invent survival percentages before enough real data exists. Until then, display “Insufficient historical data.”

13. Leaderboards

Leaderboard	Primary measurement
Seeding	Seeds reported, with monitoring/quality metrics shown separately.
Plantation	Saplings reported, with monitoring/quality metrics shown separately.
Community	Activities organized, participation and community contribution.

A future survival leaderboard can be added after enough consistent monitoring data exists. The MVP should avoid rewarding only raw quantity.

14. Monthly Goals

Admins can create monthly community goals such as seeds reported, saplings planted, participants, and monitoring records. The dashboard shows progress against each goal.

15. Species Management

Admins maintain the supported species catalogue. MVP fields may include common name, scientific name, local name, native status, basic description, basic guidance, and active/inactive status. Advanced ecological recommendations are intentionally deferred until expert review.

16. Verification & Fraud Controls

- GPS coordinates and GPS accuracy.
- Timestamp.
- Photos and optional video.
- Offline activity records with later synchronization.
- Duplicate/suspicious activity detection.
- User reports and admin moderation.

- Optional future community/trusted-volunteer verification.

The MVP should never claim that these controls provide 100% proof. They only increase confidence in the reported activity.

17. Offline Support

Offline recording is an MVP requirement because restoration activities may occur in mountains, forests, rural areas and other locations with unreliable connectivity. The app should store the activity, GPS information and media locally and synchronize when connectivity returns.

18. Notifications

- Monitoring reminder.
- Community task reminder.
- Monthly goal progress.

19. User Mobile Screens

Approximately **16 screens**: Splash, Login, Register, Forgot Password, Home, Map, Create Activity, Activity Details, Monitoring, Community, Community Task, Leaderboards, Monthly Goals, Profile, Notifications, Settings.

20. Admin Panel

Approximately **14 screens**: Dashboard, Users, User Details, Activities, Activity Details, Map, Sites, Species, Community Tasks, Goals, Reports, Monitoring, Analytics, Settings.

21. Database — Initial Entities

- users
- species
- restoration_sites
- restoration_activities
- activity_media
- monitoring_records
- community_tasks
- task_participants
- monthly_goals
- user_goal_progress
- reports
- notifications

22. Core Restoration Activity Data

restoration_activities: id, user_id, site_id, type, species_id, quantity, method, latitude, longitude, gps_accuracy, activity_date, notes, status, created_at, updated_at.

type: SEEDING or PLANTATION.

23. Core Monitoring Data

monitoring_records: id, activity_id, user_id, observed_date, observed_count, established_count, surviving_count, dead_count, condition, notes, created_at.

These fields should be validated against the final ecological monitoring methodology before production implementation.

24. Recommended Technology Stack

Layer	Technology
Mobile	Flutter
Backend	Laravel / PHP
Database	PostgreSQL + PostGIS
Python services	FastAPI — future analytics/recommendations
Maps	Google Maps, Mapbox, or OpenStreetMap-based solution after cost/licensing review
Notifications	Firebase Cloud Messaging
Media	Object storage

25. System Architecture

Flutter App → REST API → Laravel Backend → PostgreSQL/PostGIS + Media Storage + Queue

Python/FastAPI remains a separate service for future analytics, survival analysis, anomaly detection, recommendation systems and satellite-related processing. Do not over-engineer the MVP with unnecessary microservices.

26. Explicitly Out of MVP

- AI chatbot
- AI species recommendation
- Satellite analysis
- Carbon calculator/credits
- Payments
- Tree/seed marketplace
- NGO management
- Expert marketplace
- Blockchain
- Advanced gamification
- Automated species identification
- Drone integration
- IoT sensors
- Advanced weather prediction
- International expansion

27. Two-Month Delivery Plan

Period	Work
Pre-development R&D;	Problem definition, target geography, restoration methodology, species research, competitor analysis, requirements, database and API planning.
Month 1 — Weeks 1–2	UI/UX, design system, mobile flows, admin UI, database setup.
Month 1 — Weeks 3–4	Laravel APIs, authentication, activities, sites, media, goals, tasks, monitoring foundations.
Month 2 — Weeks 5–6	Flutter implementation, API integration, offline activity creation, map, photos.
Month 2 — Weeks 7–8	Monitoring, community, leaderboards, notifications, testing, bug fixing and deployment preparation.

28. MVP Acceptance Scenarios

1. A user records a seeding activity with GPS, species, quantity and photos.
2. A user records a plantation activity separately from seeding.
3. An activity can be recorded offline and synchronized later.
4. Activities appear correctly on the map.
5. Users can create and join community restoration tasks.
6. A later monitoring record can be attached to an activity.
7. Admins can create monthly goals and users can see progress.
8. Seeding and plantation leaderboards remain separate.
9. Admins can manage species and moderate reported activities.
10. The system never represents a reported activity as guaranteed survival.

29. MVP Success Definition

The MVP is successful if a real user can move through the complete loop: **record** → **organize** → **participate** → **monitor** → **learn**, while the platform keeps seeding, plantation, reported quantities, observations and actual survival outcomes clearly separated.

30. Long-Term Roadmap

Phase	Direction
Phase 1	Record + Organize + Monitor
Phase 2	Analyze — survival statistics, regional comparisons, species/method comparisons
Phase 3	Expert ecosystem — experts, NGOs, organizations, research partners
Phase 4	Intelligence — recommendations, satellite/environmental analytics, anomaly detection
Phase 5	Large-scale restoration network — communities + organizations + researchers + government

31. Final Product Statement

A community-driven ecosystem restoration platform that enables people to record and organize seeding and plantation activities, monitor their outcomes over time, and build reliable real-world data to understand what restoration approaches work best in different environments.

Review note: This is intentionally a working MVP specification. Before development, the team should review and revise the ecological terminology, monitoring definitions, species guidance, land-use warnings, privacy rules, and data model with appropriate experts.